

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper No. CHB-04A: Ancillary Chemistry-II (Basic Aspects of Chemistry)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Write down the structure of Buna-S rubber and phenol-formaldehyde resin (Bakelite).
- (b) Briefly discuss the different types of thermodynamic systems.
- (c) Draw the Lewis dot structures for the following species: (i) CO_3^{2-} and (ii) NH_4Cl .
- (d) Differentiate between binding energy and packing fraction.
- (e) Define standard enthalpy of formation (ΔH_f°).

Question 2.

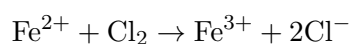
- (a) Explain the significance of the octane number and cetane number in evaluating the quality of gasoline and diesel fuels, respectively. How do they correlate with engine performance? [6]
- (b) Discuss the potential health hazards associated with the production and use of synthetic polymers. How do polymer waste disposal and recycling contribute to environment preservation? [8]

Question 3.

- (a) Predict the sign of ΔG , ΔH , and ΔS for the sublimation of solid carbon dioxide (CO_2) at room temperature. Give reasons. [4]
- (b) Assume a chemical reaction where ΔS is $+35.1 \text{ J/mol} \cdot \text{K}$ and ΔH is -2.87 kJ/mol . Calculate the change in entropy for the universe (ΔS_{univ}) for this reaction at 60°C . Predict its spontaneity. [6]
- (c) State the second law of thermodynamics. [4]

Question 4.

- (a) Standard reduction potentials of the half-reactions are given below. Predict whether the following reaction will occur spontaneously or not. Give reasons:



(Given: $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.771 \text{ V}$, $E^\circ(\text{Cl}_2/\text{Cl}^-) = +1.36 \text{ V}$). [6]

(b) Standard reduction potentials (E°) of some electrodes are: $\text{A}^+/\text{A} = -0.44 \text{ V}$, $\text{B}^{2+}/\text{B} = +0.34 \text{ V}$, and $\text{C}^+/\text{C} = +0.80 \text{ V}$. Predict the strongest oxidizing and strongest reducing agent among these. [4]

(c) Describe the construction and working of the lead storage battery, showing charging and discharging cell reactions. [4]

Question 5.

(a) Differentiate between Photosystem I and Photosystem II of photosynthesis in plants. [6]

(b) Discuss the light and dark reactions of photosynthesis. [5]

(c) State the biological role of haemoglobin and myoglobin. [3]